



East West University

Lab Report

Semester: Summer-2025

Course Title: Electronic Circuits **Course Code:** CSE251

Section: 04

Expt No: 02

Expt Name: Half-wave Diode Rectifier Circuit

Group No: 06

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Title: Half-wave Diode Rectifier Circuit

Objectives:

The primary objectives of this experiment are:

1. To understand the working principle of a half-wave rectifier.
2. To construct a half-wave rectifier circuit using a diode and resistor.
3. To observe and analyze the input alternating current (AC) and output direct current (DC) waveforms using an oscilloscope.
4. To understand how a diode rectifies an AC signal by allowing current flow in only one direction.
5. To analyze how a diode can be used to convert AC (alternating current) into DC (direct current).
6. To determine the relationship between the peak input voltage and the peak output voltage.

Equipments and Components Needed:

1. Signal Generator
2. Digital multimeter
3. Diode (1 pc)
4. Resistor $1\text{K}\Omega$ (1 pc)
5. Capacitor $10\mu\text{F}$ (1 pc)
6. Breadboard
7. Connecting wires

Theory:

A rectifier circuit converts an AC voltage with zero average into a unidirectional voltage with a non-zero average. The rectifier circuit can rectify both positive and negative half-cycles (full-wave rectifier) or only the positive half-cycle (half-wave rectifier) of a sine wave. A capacitor connected across the load resistor acts as a filter and reduces the ripple of the output voltage. The time constant of the RC network should be much larger than the period of the AC source voltage for effective filtering.

Circuit Diagram:

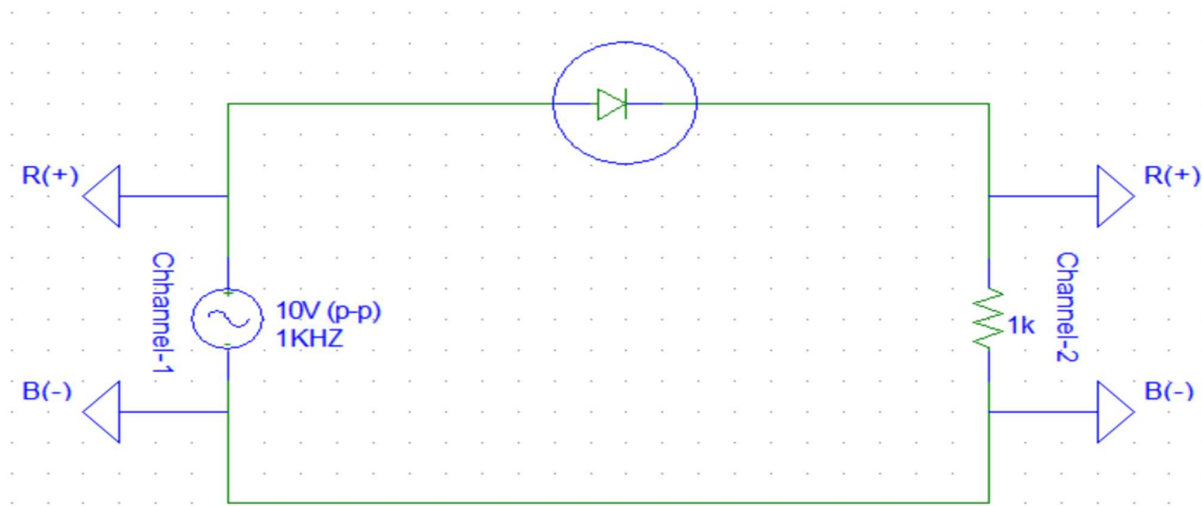


Figure-1: Set up for a half-wave diode rectifier circuit.

Pre-Lab Report Question:

1. Consider the half-wave rectifier circuit shown in Figure 1. It is fed by a 1.0 KHz sinusoid having a peak value of 5V. Design the circuit (determine the value of the parallel capacitor) so that the output will have a peak-to-peak ripple of $V_r = 0.5V$ using the formula $V_r = \frac{V_p}{fCR}$. Calculate the diode conduction time Δt and conduction angle $\omega\Delta t$ using the formula $\omega\Delta t = \cos^{-1}\left(\frac{V_p - V_r}{V_p}\right)$.

Also calculate the average and peak values of the diode currents using the formulas

$$i_{D_{avg}} = \frac{V_p - \frac{V_r}{2}}{R} \left(1 + \pi \sqrt{\frac{2V_p}{V_r}}\right) \text{ and } i_{D_{max}} = \frac{V_p - \frac{V_r}{2}}{R} \left(1 + 2\pi \sqrt{\frac{2V_p}{V_r}}\right)$$

Answer

Given,

Input frequency, $f = 1.0 \text{ kHz} = 1000$

Peak voltage, $V_p = 5 \text{ V}$

Load resistor, $R = 1 \text{ k}\Omega = 1000 \Omega$

Desired ripple voltage, $V_r = 0.5 \text{ V}$

$$V_r = \frac{V_p}{fCR}$$

$$\therefore \text{Parallel capacitor, } C = \frac{V_p}{fRV_r} = \frac{5}{1000 \times 1000 \times 0.5} = 10 \times 10^{-6} \text{ F} = 10 \mu\text{F}$$

diode conduction angle, $\omega\Delta t = \cos^{-1}\left(\frac{V_p - V_r}{V_p}\right) = \cos^{-1}\left(\frac{5 - 0.5}{5}\right) = 0.451 \text{ rad}$

We know, $\omega = 2\pi f \text{ rad/s}$

$\therefore$ Diode conduction time, $\Delta t = \frac{\omega\Delta t}{\omega} = \frac{\omega\Delta t}{2\pi f} = \frac{0.451}{2\pi \times 1000} = 0.072 \text{ ms}$

=

$$i_{D_{avg}} = \frac{V_p - \frac{V_r}{2}}{R} \left(1 + \pi \sqrt{\frac{2V_p}{V_r}} \right) = \frac{5 - \frac{0.5}{2}}{1000} \left(1 + \pi \sqrt{\frac{2 \times 5}{0.5}} \right) = 71.45 \text{ mA}$$

$$i_{D_{max}} = \frac{V_p - \frac{V_r}{2}}{R} \left(1 + 2\pi \sqrt{\frac{2V_p}{V_r}} \right) = \frac{5 - \frac{0.5}{2}}{1000} \left(1 + 2\pi \sqrt{\frac{2 \times 5}{0.5}} \right) = 138.22 \text{ mA}$$

Experimental Data:

$R = 0.987 \text{ K}\Omega$

Peak Value, $\Delta V_p = 0.6767 \text{ V}$

Time, $\Delta t = 0.2 \text{ ms}$

Ripple Voltage, $V_r = 0.6968 \text{ V}$

Average Voltage, $V_o = 12.87 \text{ V}$

Post-Lab Report Questions:

1. Compare the measured value of ΔV_p with the built-in voltage of Expt 1.

Answer

Measure value of Experiment-1

$\Delta V_p = 0.6767 \text{ V}$

Which is close to the $\Delta V_p = 0.7 \text{ V}$ for silicon diode

Comparison table:

Measure value of Experiment-1	Built-in voltage of silicon diode	Discrepancy
0.6767 V	0.7 V	3.3%

2. Compare your measured Δt with your pre-lab value and make a comment.

Answer

Measured Value, $\Delta t = 0.2$ ms

Calculated pre-lab value, $\Delta t = 0.072$ ms

Comparison table:

Measure value Δt	Calculated pre-lab value Δt	Discrepancy
0.2 ms	0.072 ms	64%

Comment: There is a little bit difference between Measured Δt and calculated pre-lab Δt

3. Calculate the peak-to-peak ripple voltage from the formula $\omega\Delta t = \sqrt{\frac{2V_r}{V_p}}$ and compare it with your measured data and pre-lab data and make a comment.

Answer

Input frequency, $f = 1.0$ kHz = 1000

Peak voltage, $V_p = 0.6767$ V

Measured Value, $\Delta t = 0.2$ ms

Measure value $V_{r_1} = 0.6968$ V

Pre-lab value $V_{r_2} = 0.5$ V

$$\omega\Delta t = \sqrt{\frac{2V_r}{V_p}}$$

$$\therefore \text{Peak-to-peak ripple voltage } V_r = \frac{V_p \times (\omega\Delta t)^2}{2} = \frac{V_p \times (2\pi f \times \Delta t)^2}{2} = \frac{0.6767 \times (2\pi \times 1000 \times 0.2)^2}{2} \\ = 0.5343 \text{ V}$$

Comparison table:

Calculated value	Measure value	Pre-lab value
0.5343 V	0.6968 V	0.5 V

Comment: There is a little bit difference between calculated, measured value and prelab values.

4. Calculate the average output voltage from $V_o = V_p - \frac{V_r}{2}$ and compare it with the measurement.

Answer

$$V_p = 0.6767 \text{ V}$$

$$V_r = 0.6968 \text{ V}$$

$$\therefore \text{Average output voltage from } V_o = V_p - \frac{V_r}{2} = 0.6767 - \frac{0.6968}{2} = 0.3283 \text{ V}$$

Comparison table:

Measure value V_o	Calculated value V_o
12.87 V	0.3283 V

5. With $I_L = \frac{V_o}{R}$ calculate the average and maximum diode currents using the formulas

$$i_{D_{avg}} = I_L \left(1 + \pi \sqrt{\frac{2V_p}{V_r}} \right) \text{ and } i_{D_{max}} = I_L \left(1 + 2\pi \sqrt{\frac{2V_p}{V_r}} \right) \text{ and measured value of } V_r, \text{ and}$$

compare with your pre-lab values and make a comment.

Answer

$$I_L = \frac{V_o}{R} = \frac{12.87}{0.987} = 13.16 \text{ mA}$$

$$i_{D_{avg}} = I_L \left(1 + \pi \sqrt{\frac{2V_p}{V_r}} \right) = 13.16 \times \left(1 + \pi \times \sqrt{\frac{2 \times 0.6767}{0.6968}} \right) = 70.78 \text{ mA}$$

$$i_{D_{max}} = I_L \left(1 + 2\pi \sqrt{\frac{2V_p}{V_r}} \right) = 13.16 \times \left(1 + 2\pi \times \sqrt{\frac{2 \times 0.6767}{0.6968}} \right) = 128.34$$

Comparison table:

	Measure value	Calculated pre-lab value	Discrepancy
$i_{D_{avg}}$	70.78 mA	71.45 mA	0.938%
$i_{D_{max}}$	128.34 mA	138.22 mA	7.148%

6. Simulate the half-wave rectifier circuit in PSPICE for $C = 10\ \mu\text{F}$ and submit the input and output plots (on same graph). Use transient analysis of PSPICE for 4 cycles of input (4 ms). Modify the diode parameters following the same procedure and the same parameters values used in Expt 1.

Answer

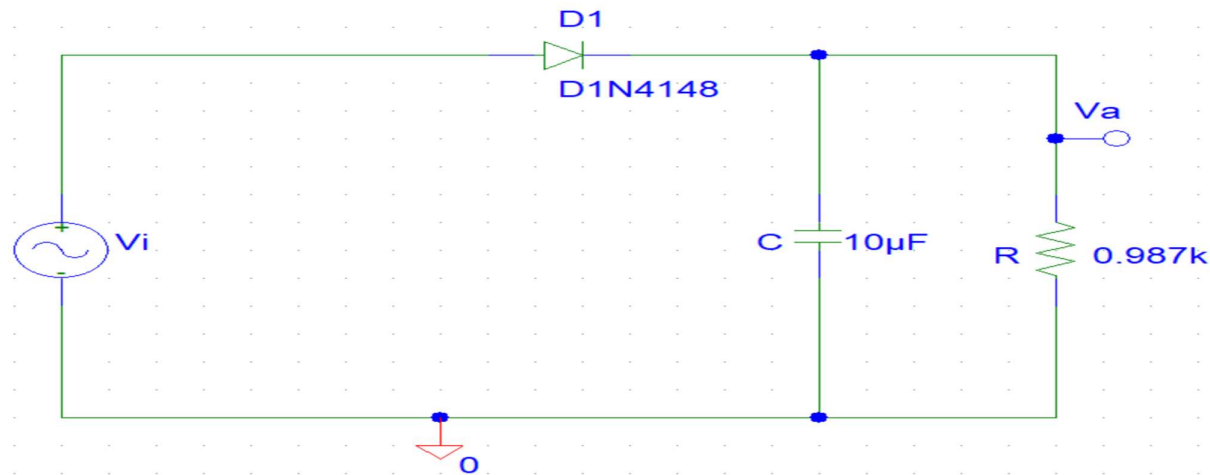


Figure-2

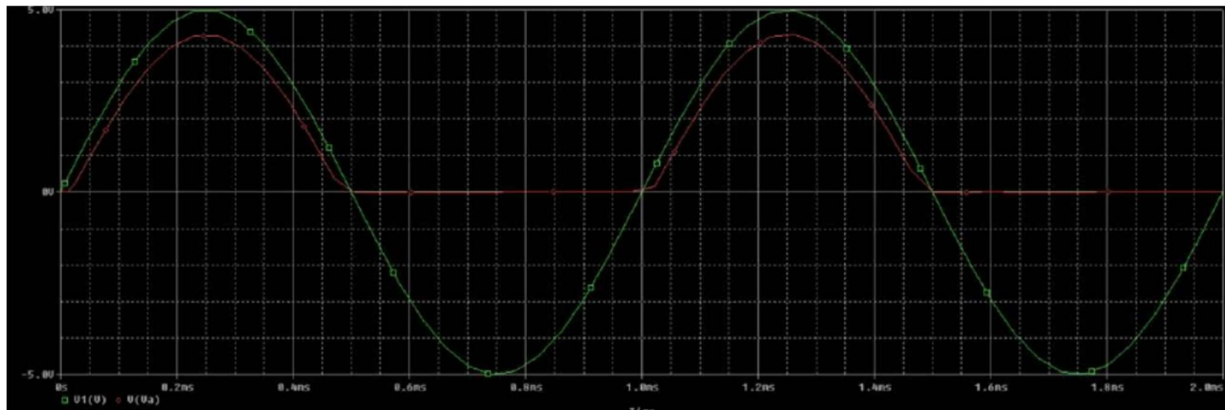


Figure-3

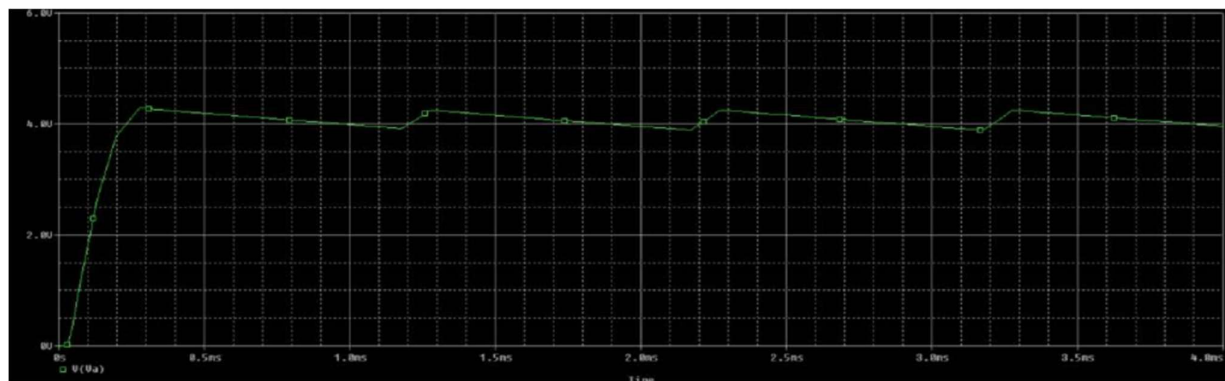


Figure-4

Discussion: The experiment successfully studied how a half wave diode rectifier works, focusing on its basic principles, circuit construction, and the relationship between input and output voltages.

The half-wave rectifier circuit demonstrated the fundamental behavior of a p-n junction diode in converting alternative current (AC) into a unidirectional (pulsating DC) current with a non-zero average using a simple diode. The half-wave rectifier only processes one half of the sine wave. Adding a capacitor across the load resistor helps smooth out the V_r , ~~at V_r~~ . The measured ΔV_r was close to the silicon diode, but a noticeable difference was found in Δt and V_o .

Prospice simulation showed the input AC signal and the rectified output with reduced ripple. However, discrepancies were found in V_r and waveform shape. However, i_{Davg} and i_{Dmax} were found consistent.

Future work could involve using more precise measurement instruments, conducting multiple trials to reduce random errors, and exploring different diode types or filter configurations to further analyze their impact on rectifier performance.